S4G Image Workflow Documentation

Overview

The workflow downloads S4G FITS images, builds a geometry manifest, and plots diagnostic isophote PDFs.

Workflow:

1. download_s4g_images.py
   Downloads each galaxy's S4G 3.6 micron FITS image from IRSA.
   Output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\s4g_images_36um
2. build_s4g_geometry_manifest.py
   Combines the downloaded FITS image paths with local and catalogue geometry data.
   Output: s4g_image_downloader\geometry_output\s4g_image_geometry_manifest.csv
3. plot_s4g_isophote_axes.py
   Reads the manifest and FITS images, then creates major/minor-axis isophote diagnostic PDFs.
   Output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\isophote_output

Function-level overview: download_s4g_images.py

- parse_args: Defines command-line options such as --limit, --dry-run, --outdir, and --timeout.
- read_galaxy_names: Reads scrambled_map.txt and returns the galaxy names from the third column.
- download_image: Builds the IRSA FITS URL, skips existing files, optionally dry-runs, and saves valid downloads.

- main: Resolves paths, creates the output directory, reads the galaxy list, and downloads each requested image.

Function-level overview: build_s4g_geometry_manifest.py

- clean_value: Converts table text to numbers and normalizes known missing or invalid values to None.
- read_scrambled_map: Loads scrambled index, original index, and galaxy name from scrambled_map.txt.
- read_s4g_table: Reads local s4gbars_table.dat and keeps cleaned S4G bar and galaxy fields by name.
- table_value: Converts Astropy masked, scalar, or byte values into ordinary Python values.
- read_cached_or_fetch_table: Loads a VizieR catalogue from cache or downloads it and writes an ECSV cache file.
- keyed_table: Turns a catalogue table into a dictionary keyed by a selected name column.
- keyed_herrera_bars: Filters Herrera-Endoqui rows to bar entries and keys them by galaxy name.
- fetch_catalog_geometry: Fetches or loads Herrera-Endoqui, Salo, and Diaz-Garcia geometry catalogues.
- fits_metadata: Reads FITS header metadata needed for image size, pixel scale, centre, and orientation.
- build_manifest: Creates one manifest row per galaxy, merging local tables, catalogue geometry, FITS metadata, and notes.
- write_manifest: Writes the manifest rows to CSV using the fixed OUTPUT_FIELDS column order.
- parse_args: Defines paths and catalogue-cache switches such as --image-dir, --no-vizier, and --refresh-cache.

- main: Loads optional catalogue geometry, builds the manifest, writes the CSV, and prints image-link counts.

Function-level overview: plot_s4g_isophote_axes.py

- parse_float: Safely converts manifest string values to finite floats.
- safe_filename: Creates filesystem-safe PDF names from galaxy names.
- read_manifest: Reads the geometry manifest CSV into dictionaries.
- pa_endpoint: Converts a position angle and radius into x/y endpoints for drawing or sampling lines.
- draw_pa_line: Draws major or minor axis lines on the image panel.
- extract_centered_subimage: Crops a centre-based image cutout and builds arcsecond coordinate arrays.
- profile_at_pa: Samples a smoothed intensity profile along a requested position angle.
- robust_log_image: Creates a stable log-scaled image and contour levels even when pixels include zeros or NaNs.
- deprojected_profile_radius: Uses angle_utils.deprojectr to convert projected profile radii to deprojected radii.
- required_geometry: Pulls required geometry fields from a manifest row and rejects incomplete rows.
- make_plot: Creates the two-panel diagnostic plot for a single galaxy and writes it to individual and/or combined PDFs.
- selected_rows: Applies --names and --limit filters to the manifest rows.
- parse_args: Defines plotting options such as --manifest, --image-dir, --output-dir, --names, and --no-combined.
- main: Creates output folders, iterates through selected galaxies, writes PDFs, and reports skipped cases.

Program detail: download_s4g_images.py

Purpose: downloads S4G 3.6 micron FITS images from IRSA for galaxies listed in scrambled_map.txt.

Default output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\s4g_images_36um

Inputs: the scrambled map file supplies the galaxy names from its third column.

Behaviour: existing FITS files are skipped, --dry-run prints planned downloads, and --limit supports small test batches.

Key options: --galaxy-list, --outdir, --base-url, --limit, --timeout, and --dry-run.

Key commands: python download_s4g_images.py --dry-run --limit 5; python download_s4g_images.py --limit 1; python download_s4g_images.py

Program detail: build_s4g_geometry_manifest.py

Purpose: builds a CSV manifest that links each galaxy to FITS metadata and geometry measurements.

Default output: s4g_image_downloader\geometry_output\s4g_image_geometry_manifest.csv

Default image folder: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\s4g_images_36um

Default cache folder: s4g_image_downloader\geometry_catalog_cache.

Inputs: scrambled_map.txt, s4gbars_table.dat, downloaded FITS headers, and optional VizieR catalogues.

Catalogue sources: Herrera-Endoqui+2015, Salo+2015, and Diaz-Garcia+2016.

Output fields include image_exists, image_path, pixel scale, CRPIX/CRVAL, disc geometry, bar geometry, stellar mass, and notes.

Key options: --scrambled-map, --s4g-table, --image-dir, --cache-dir, --no-vizier, --refresh-cache, and --output-csv.

Key commands: python build_s4g_geometry_manifest.py; python build_s4g_geometry_manifest.py --no-vizier; python build_s4g_geometry_manifest.py --refresh-cache

Program detail: plot_s4g_isophote_axes.py

Purpose: creates Figure-1-style diagnostic PDFs for each selected galaxy.

Default output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\isophote_output

Default manifest: s4g_image_downloader\geometry_output\s4g_image_geometry_manifest.csv.

Default image fallback folder: D:\Dropbox\Public Documents\UCLAN\MSc Research\Erwin\s4g_images_36um.

Inputs: the geometry manifest and the downloaded FITS images.

Plot contents: log-scaled S4G isophotes, observed bar major axis, projected minor axis, and major/minor-axis intensity cuts.

Behaviour: incomplete rows or missing FITS images are skipped and reported after the batch.

Key options: --manifest, --image-dir, --output-dir, --combined-pdf, --limit, --names, --profile-width, --no-individual, and --no-combined.

Key commands: python plot_s4g_isophote_axes.py; python plot_s4g_isophote_axes.py --limit 3; python plot_s4g_isophote_axes.py --names NGC1879 IC0600

Recommended run order:

1. python download_s4g_images.py --dry-run --limit 5
2. python download_s4g_images.py --limit 5
3. python build_s4g_geometry_manifest.py --no-vizier
4. python plot_s4g_isophote_axes.py --limit 3